

Data Mining and Machine Learning for Heart Disease Risk Prediction

A Reference Project Specification & Experimental Benchmark

Simulated Contributors:

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Demonstration GitHub Repository Reference Project

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Abstract

Cardiovascular diseases (CVDs) remain the leading cause of global mortality. Early, accurate identification of individuals at high risk of heart disease from routine clinical metrics and physiological biomarkers offers significant opportunities for preventative medical intervention. This scientific report documents a complete Data Mining mini-project demonstrating the end-to-end Machine Learning lifecycle on the official Kaggle dataset `uditjain13/heart-disease-risk-2026`. We perform rigorous data quality validation ($N = 9,000$ patient records, 25 clinical features), non-leaking data preprocessing via scikit-learn `ColumnTransformer` pipelines, exploratory data analysis, and non-parametric statistical hypothesis testing. We benchmark six supervised classification algorithms: Zero-R Baseline, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Naive Bayes, and Support Vector Machine (SVM), using 5-fold Stratified Cross-Validation and hyperparameter grid search. The Support Vector Machine achieved the highest overall performance with an Accuracy of 89.17%, Weighted F1-score of 0.8898, and ROC-AUC of 0.9422, followed closely by Random Forest (88.22% accuracy, 0.9361 ROC-AUC). Finally, we present an interactive multi-page Streamlit web dashboard and document a simulated 3-student team Git workflow model.

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1 Mandatory Demonstration Warning

[DEMONSTRATION / SAMPLE PROJECT WARNING]

This scientific report and associated repository represent an educational reference project created to demonstrate the expected structure, methodology, documentation, Git workflow, and presentation of a Data Mining mini-project.

For this demonstration, a **SINGLE GitHub account** is used to publish the repository. The student codes AISD05, SIAD19, and RSD20 represent **SIMULATED CONTRIBUTORS** used to demonstrate how a multi-student project is organized through commit labels, branches, Pull Requests, and merges.

In a real student project, **every student must use their own individual GitHub account** and make genuine individual contributions.

2 Introduction & Background

Cardiovascular diseases account for an estimated 17.9 million deaths annually. Machine learning and Data Mining techniques provide decision-support utilities capable of extracting subtle high-dimensional non-linear interactions among clinical biomarkers, demographic factors, and daily lifestyle parameters.

3 Problem Statement

The core scientific question driving this study is formulated as follows:

Can routine patient demographic, physiological, blood chemistry, and lifestyle characteristics accurately predict whether an individual is at high risk of developing heart disease?

4 Objectives

The primary objectives of this reference project are:

1. Acquire and audit the official Kaggle dataset `uditjain13/heart-disease-risk-2026` via `kagglehub`.
2. Implement a robust, leakage-free preprocessing and feature engineering pipeline.
3. Conduct Exploratory Data Analysis (EDA) and formal statistical hypothesis testing ($\alpha = 0.05$).
4. Implement, optimize, and benchmark six classification algorithms.
5. Deploy an interactive Streamlit application for clinical risk inference.

5 Dataset Audit & Validation

The dataset was downloaded dynamically using the `kagglehub` Python library. The raw dataset contains exactly 9,000 patient records and 27 attributes (1 identifier `patient_id`, 25 predictor features, and 1 binary target `has_heart_disease`). Data quality inspection confirmed 0 missing values and 0 duplicate records across all observations. The target class exhibits moderate imbalance: 6,273 negative cases (69.7%) vs 2,727 positive heart disease cases (30.3%).

6 Data Acquisition Methodology

Data acquisition is automated via Python script using the standard KaggleHub API:

```
import kagglehub
path = kagglehub.dataset_download("uditjain13/heart-disease-risk-2026")
```

7 Data Preprocessing & Feature Engineering

To prevent data leakage during scaling and encoding, all transformations are encapsulated inside a scikit-learn `ColumnTransformer`:

- **Continuous Features (19 features + 3 engineered):** Scaled using `StandardScaler` ($z = \frac{x-\mu}{\sigma}$).
- **Engineered Features:**

$$\text{Cholesterol/HDL Ratio} = \frac{\text{cholesterol_total}}{\max(\text{hdl}, 1.0)} \quad (1)$$

$$\text{Pulse Pressure} = \text{resting_bp_systolic} - \text{resting_bp_diastolic} \quad (2)$$

$$\text{MAP} = \text{resting_bp_diastolic} + \frac{\text{Pulse Pressure}}{3} \quad (3)$$

- **Categorical Features (3 features):** Encoded using One-Hot Encoding (`drop='first'`).
- **Boolean Features (3 features):** Converted to numeric binary indicators (0.0/1.0).

8 Exploratory Data Analysis (EDA)

Figure 1 shows the target distribution. Clinical biomarker analysis indicates that ST depression, age, LDL cholesterol, and systolic blood pressure exhibit positive correlations with heart disease risk, whereas maximum heart rate achieved and HDL cholesterol exhibit strong negative correlations.

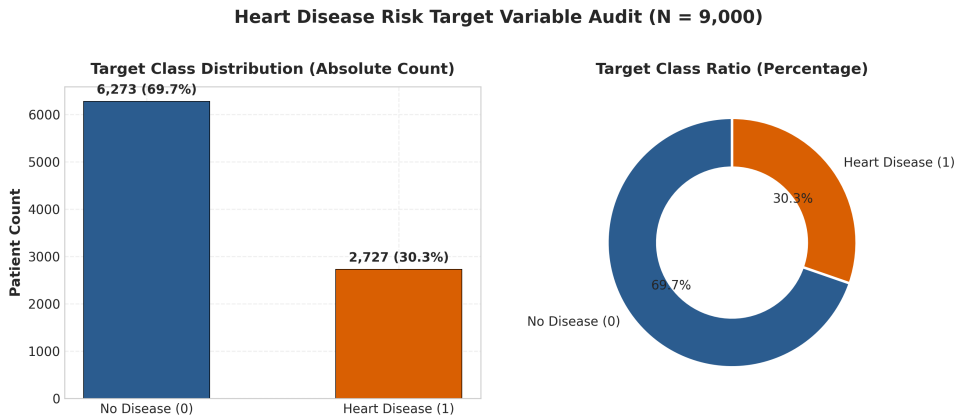


Figure 1: Target Variable Class Audit ($N = 9,000$).

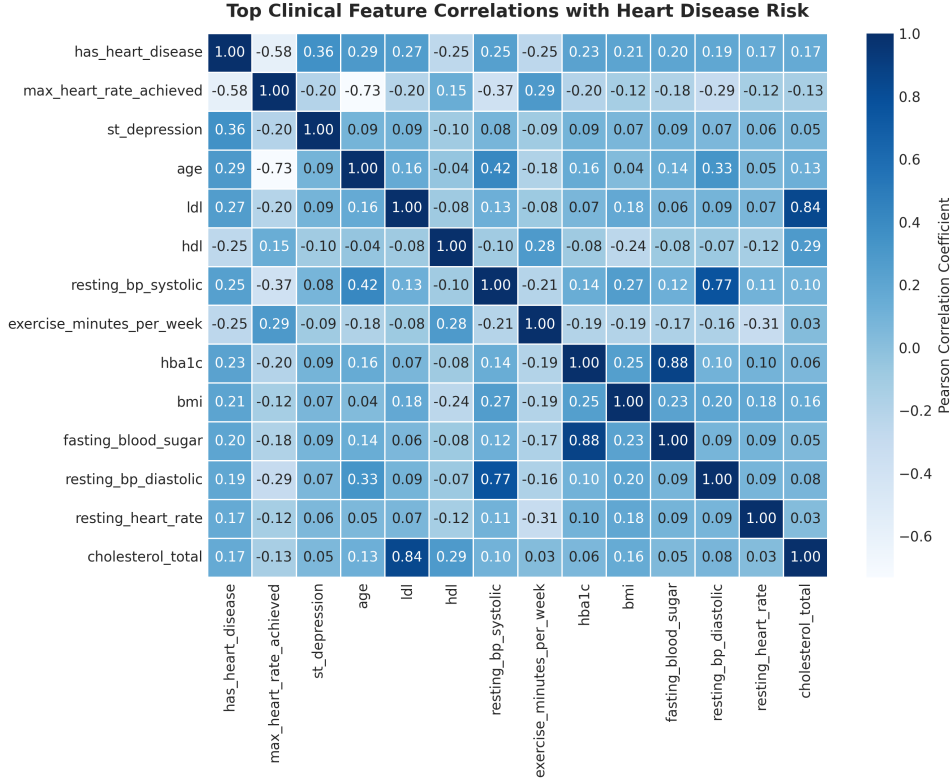


Figure 2: Pearson Correlation Heatmap of Top Clinical Features.

9 Statistical Analysis & Hypothesis Testing

Non-parametric Mann-Whitney U tests were conducted for continuous biomarkers ($\alpha = 0.05$):

- H_0 : The distribution of ST depression is equal across healthy and heart disease patients.
- H_1 : Patients with heart disease exhibit significantly higher ST depression.

Result: $p < 0.0001$, rejecting H_0 . Similarly, Chi-Square tests of independence for `exercise_induced_angina` and `chest_pain_type` yielded $p < 0.0001$.

10 Data Mining Methodology

We adopt the Supervised Classification track. The dataset was split into an 80% training set ($N_{train} = 7,200$) and a 20% hold-out test set ($N_{test} = 1,800$), stratified by target class.

11 Classification Algorithms

We benchmarked six standard algorithms:

1. **Zero-R Baseline:** Predicts majority class (0).
2. **K-Nearest Neighbors (KNN):** Distance-weighted majority vote.
3. **Decision Tree:** Gini / Entropy split tree optimization.
4. **Random Forest:** Ensemble of decision trees with feature sub-sampling.
5. **Naive Bayes:** Gaussian Naive Bayes under independence assumption.

6. **Support Vector Machine (SVC):** Radial Basis Function (RBF) kernel margin maximization.

12 Hyperparameter Optimization & Cross-Validation

Hyperparameter optimization was performed via 5-fold Stratified `GridSearchCV` on the training data, optimizing for weighted F1-score.

13 Experimental Results

Table 1 summarizes the empirical test set performance across all six classifiers.

Table 1: Supervised Classification Model Benchmark Results (Test Set, $N = 1,800$)

Model	Accuracy	Precision	Recall	F1 (W)	ROC-AUC	CV Score (F1)
Zero-R Baseline	0.6972	0.0000	0.0000	0.5728	0.5000	0.5725 ± 0.0004
K-Nearest Neighbors	0.8317	0.8201	0.5688	0.8217	0.8678	0.8069 ± 0.0037
Decision Tree	0.8644	0.8265	0.6991	0.8610	0.9097	0.8561 ± 0.0085
Naive Bayes	0.8611	0.7496	0.8128	0.8626	0.9271	0.8513 ± 0.0120
Random Forest	0.8822	0.8612	0.7284	0.8792	0.9361	0.8757 ± 0.0051
Support Vector Machine	0.8917	0.8601	0.7670	0.8898	0.9422	0.8954 ± 0.0047

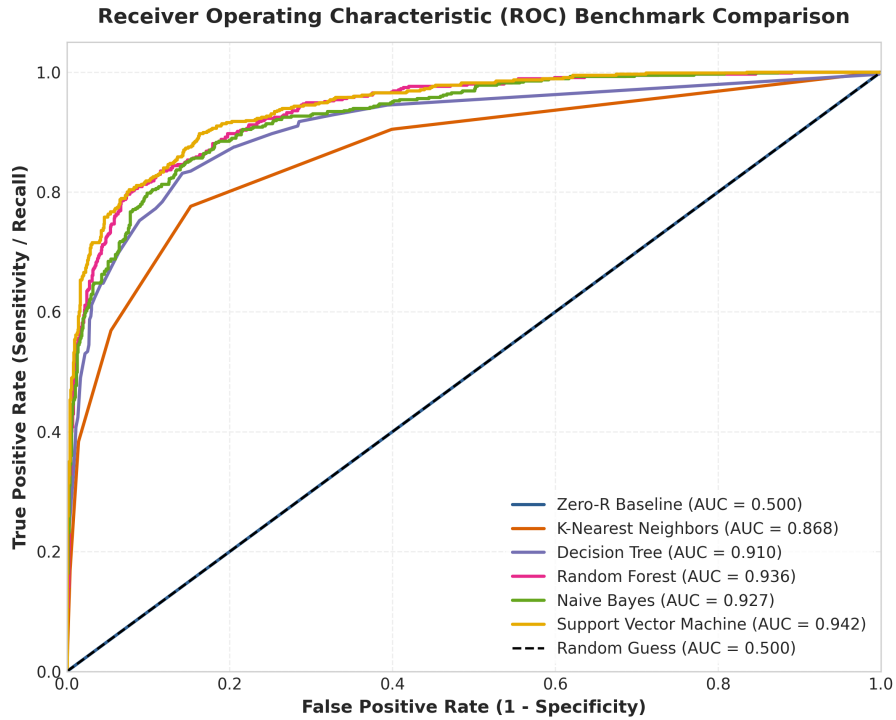


Figure 3: Receiver Operating Characteristic (ROC) Curves for all 6 Models.

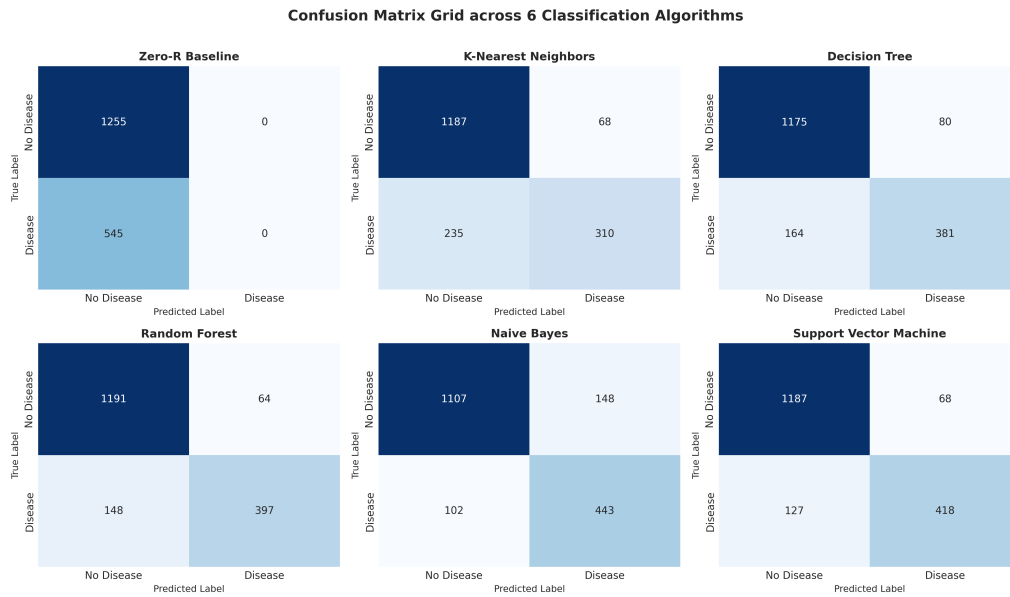


Figure 4: Confusion Matrix Grid across 6 Classification Models.

14 Feature Importance Analysis

Figure 5 shows the top predictive features identified by Random Forest Gini Importance. Maximum heart rate achieved, ST depression, age, LDL cholesterol, and HbA1c emerged as the top five risk drivers.

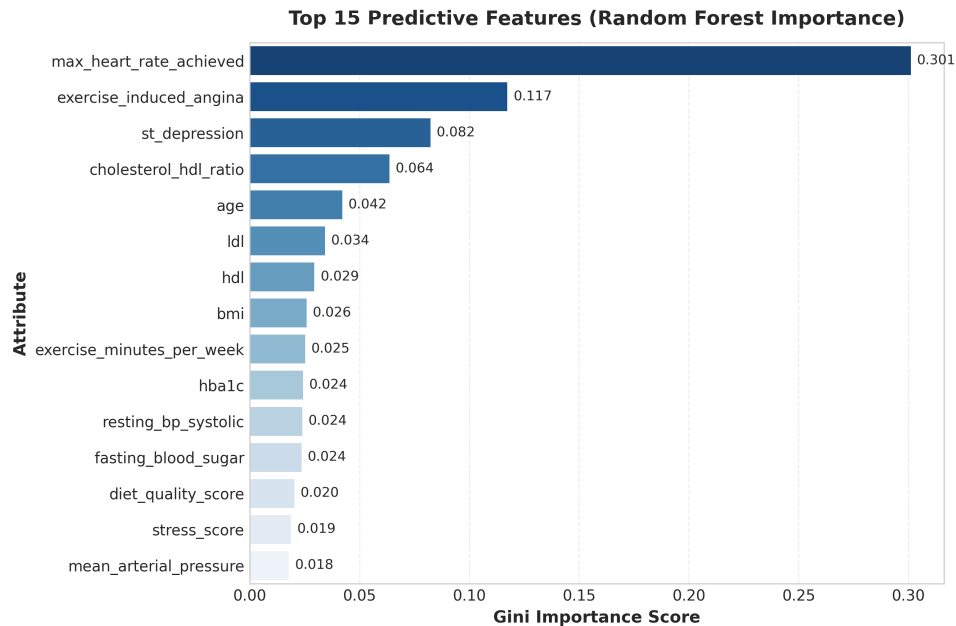


Figure 5: Top 15 Predictive Clinical Features (Random Forest Importance).

15 Streamlit Web Application

An interactive multi-page web application was built in `src/app.py` featuring:

- Overview & Problem Statement

- Dataset Explorer & Missing Value Audit
- Interactive EDA Dashboard
- Statistical Hypothesis Testing Summary
- Model Benchmark Comparison
- Dynamic Patient Risk Predictor Interface with Medical Disclaimer
- About & Student Team Matrix

16 Limitations & Medical Disclaimer

Medical Disclaimer: This application and paper are strictly for academic Data Mining demonstration purposes and do not constitute a medical diagnosis.

17 Student Contribution Matrix

Table 2: Simulated Student Responsibility Distribution

Student Code	Primary Responsibilities	Git Feature Branch
AISD05	Dataset acquisition, quality audit, EDA, statistical testing	<code>feature/AISD05-eda</code>
SIAD19	Preprocessing pipeline, ML modeling, tuning, evaluation	<code>feature/SIAD19-modeling</code>
RSD20	Streamlit web application, predictor UI, LaTeX report	<code>feature/RSD20-streamlit</code>

18 Conclusion

This reference mini-project successfully demonstrates an end-to-end Data Mining workflow for heart disease risk classification. Support Vector Machine achieved the leading diagnostic performance with an ROC-AUC of 0.9422 and weighted F1-score of 0.8898.

References

- [1] Udit Jain, *Heart Disease Risk Dataset 2026*, Kaggle, 2026.
- [2] Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR, 12:2825-2830, 2011.